Case Study: Patient Management System Using Java Stream API

Problem Statement

Consider a healthcare system that maintains a collection of Patient objects. Each Patient object contains the following attributes:

1. **Patient ID** (patient_id)
2. **Patient Name** (patient_name)
3. **Department** (dept) – The department to which the patient belongs (e.g., Cardiology, Neurology, Orthopedics).
4. **Doctor Name** (doctor_name)
5. **Diagnosis** (diagnosis)
6. **Treatment Given** (treatment_given)
7. **Patient Type** (patient_type) – ICU, Outward, or Inward.
8. **Heart Rate** (heart_rate)
9. **Blood Pressure** (BP)
10. **Sugar Value** (sugar_value)
11. **Severity** (severity) – A value ranging from 1 to 5.

Tasks

1. Find All ICU Patients

Retrieve and display all patients whose patient type is ICU.

2. Find ICU Patients with Elevated Heart Rate

Retrieve and display all ICU patients whose heart rate is greater than 100.

3. Sort Patients by Severity

Sort and display all patients based on their severity level in descending order (highest severity first).

Implementation Requirements

- Create two separate Java classes: Patient.java and Main.java.
- Use a collection to store Patient objects.
- Perform all three operations using the Java Stream API.
- Display the required patient details for each operation.